

Models

Model	k	Starting belief	Learning rate	Choice rule	Stickiness	β by last outcome	Free parameters	What it assumes	Median fitted values	Δ BIC (summed)
M6	3	pre-task rating	reciprocate vs. defect	belief: $\beta(Q - 0.5)$	no	no	α_r , α_d , β	Starts at the pre-task stereotype rating; separate learning rates after reciprocation and defection; chooses by belief (stakes ignored).	$\alpha_r = 0.09$, $\alpha_d = 0.09$, $\beta = 4.21$	0
M7b	4	pre-task rating	reciprocate vs. defect	belief: $\beta(Q - 0.5)$	yes	no	α_r , α_d , β , ϕ	Starts at the pre-task stereotype rating; separate learning rates after reciprocation and defection; chooses by belief (stakes ignored); tends to repeat its last choice.	$\alpha_r = 0.10$, $\alpha_d = 0.09$, $\beta = 3.74$, $\phi = 0.19$	100
M8	5	pre-task rating	reciprocate vs. defect	belief: $\beta(Q - 0.5)$	yes	yes	α_r , α_d , β_r , β_d , ϕ	Starts at the pre-task stereotype rating; separate learning rates after reciprocation and defection; chooses by belief (stakes ignored); tends to repeat its last choice; choice consistency depends on the last outcome.	$\alpha_r = 0.09$, $\alpha_d = 0.10$, $\beta_r = 3.64$, $\beta_d = 3.95$, $\phi = 0.18$	228
M6v	3	pre-task rating	reciprocate vs. defect	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	no	no	α_r , α_d , β	Starts at the pre-task stereotype rating; separate learning rates after reciprocation and defection; chooses	$\alpha_r = 0.21$, $\alpha_d = 0.04$, $\beta = 0.63$	241

Model	k	Starting belief	Learning rate	Choice rule	Stickiness	β by last outcome	Free parameters	What it assumes	Median fitted values	Δ BIC (summed)
								by expected value (stakes matter).		
M7	3	pre-task rating	single	belief: $\beta(Q - 0.5)$	yes	no	alpha, beta, phi	Starts at the pre-task stereotype rating; one learning rate for everything; chooses by belief (stakes ignored); tends to repeat its last choice.	alpha = 0.11, beta = 2.61, phi = 0.39	249
M7bv	4	pre-task rating	reciprocate vs. defect	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	yes	no	alpha_r, alpha_d, beta, phi	Starts at the pre-task stereotype rating; separate learning rates after reciprocation and defection; chooses by expected value (stakes matter); tends to repeat its last choice.	alpha_r = 0.21, alpha_d = 0.05, beta = 0.54, phi = 0.36	285
M2	3	0.5	reciprocate vs. defect	belief: $\beta(Q - 0.5)$	no	no	alpha_r, alpha_d, beta	Starts at 0.5; separate learning rates after reciprocation and defection; chooses by belief (stakes ignored).	alpha_r = 0.10, alpha_d = 0.04, beta = 4.54	307
M3b	4	0.5	reciprocate vs. defect	belief: $\beta(Q - 0.5)$	yes	no	alpha_r, alpha_d, beta, phi	Starts at 0.5; separate learning rates after reciprocation and defection; chooses by belief (stakes ignored); tends to repeat its last choice.	alpha_r = 0.10, alpha_d = 0.05, beta = 4.07, phi = 0.29	338

Model	k	Starting belief	Learning rate	Choice rule	Stickiness	β by last outcome	Free parameters	What it assumes	Median fitted values	Δ BIC (summed)
M5	2	pre-task rating	single	belief: $\beta(Q - 0.5)$	no	no	alpha, beta	Starts at the pre-task stereotype rating; one learning rate for everything; chooses by belief (stakes ignored).	alpha = 0.11, beta = 3.55	392
M7v	3	pre-task rating	single	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	yes	no	alpha, beta, phi	Starts at the pre-task stereotype rating; one learning rate for everything; chooses by expected value (stakes matter); tends to repeat its last choice.	alpha = 0.13, beta = 0.19, phi = 0.50	480
M4	5	0.5	reciprocate vs. defect	belief: $\beta(Q - 0.5)$	yes	yes	alpha_r, alpha_d, beta_r, beta_d, phi	Starts at 0.5; separate learning rates after reciprocation and defection; chooses by belief (stakes ignored); tends to repeat its last choice; choice consistency depends on the last outcome.	alpha_r = 0.11, alpha_d = 0.04, beta_r = 4.36, beta_d = 5.44, phi = 0.28	490
M3bv	4	0.5	reciprocate vs. defect	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	yes	no	alpha_r, alpha_d, beta, phi	Starts at 0.5; separate learning rates after reciprocation and defection; chooses by expected value (stakes matter); tends to repeat its last choice.	alpha_r = 0.43, alpha_d = 0.06, beta = 0.42, phi = 0.42	509
M_cong_SP	5	pre-task rating	by congruence (3)	belief: $\beta(Q - 0.5)$	yes	no	alpha_c, alpha_i, alpha_amb, beta, phi	Starts at the pre-task stereotype rating; separate learning rates for congruent, incongruent, and	alpha_c = 0.11, alpha_i = 0.07, alpha_amb = 0.10, beta = 3.11, phi = 0.42	528

Model	k	Starting belief	Learning rate	Choice rule	Stickiness	β by last outcome	Free parameters	What it assumes	Median fitted values	Δ BIC (summed)
								ambiguous partners; chooses by belief (stakes ignored); tends to repeat its last choice.		
M3	3	0.5	single	belief: $\beta(Q - 0.5)$	yes	no	alpha, beta, phi	Starts at 0.5; one learning rate for everything; chooses by belief (stakes ignored); tends to repeat its last choice.	alpha = 0.35, beta = 1.32, phi = 0.53	542
M2v	3	0.5	reciprocate vs. defect	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	no	no	alpha_r, alpha_d, beta	Starts at 0.5; separate learning rates after reciprocation and defection; chooses by expected value (stakes matter).	alpha_r = 0.40, alpha_d = 0.06, beta = 0.48	584
M3v	3	0.5	single	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	yes	no	alpha, beta, phi	Starts at 0.5; one learning rate for everything; chooses by expected value (stakes matter); tends to repeat its last choice.	alpha = 0.43, beta = 0.08, phi = 0.58	617
M_cong_S	4	pre-task rating	by congruence (3)	belief: $\beta(Q - 0.5)$	no	no	alpha_c, alpha_i, alpha_amb, beta	Starts at the pre-task stereotype rating; separate learning rates for congruent, incongruent, and ambiguous partners; chooses by belief (stakes ignored).	alpha_c = 0.11, alpha_i = 0.07, alpha_amb = 0.07, beta = 4.33	638
M5v	2	pre-task rating	single	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	no	no	alpha, beta	Starts at the pre-task stereotype rating; one learning rate for everything; chooses by	alpha = 0.07, beta = 0.24	765

Model	k	Starting belief	Learning rate	Choice rule	Stickiness	β by last outcome	Free parameters	What it assumes	Median fitted values	Δ BIC (summed)
								expected value (stakes matter).		
M_cong_SPv	5	pre-task rating	by congruence (3)	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	yes	no	alpha_c, alpha_i, alpha_amb, beta, phi	Starts at the pre-task stereotype rating; separate learning rates for congruent, incongruent, and ambiguous partners; chooses by expected value (stakes matter); tends to repeat its last choice.	alpha_c = 0.10, alpha_i = 0.08, alpha_amb = 0.07, beta = 0.28, phi = 0.51	805
M_cong_P	5	0.5	by congruence (3)	belief: $\beta(Q - 0.5)$	yes	no	alpha_c, alpha_i, alpha_amb, beta, phi	Starts at 0.5; separate learning rates for congruent, incongruent, and ambiguous partners; chooses by belief (stakes ignored); tends to repeat its last choice.	alpha_c = 0.15, alpha_i = 0.02, alpha_amb = 0.02, beta = 3.22, phi = 0.44	815
M0	0	—	none	P = 0.5	no	no	—	Null: random 50/50 choice, no learning.	—	881
M1	2	0.5	single	belief: $\beta(Q - 0.5)$	no	no	alpha, beta	Starts at 0.5; one learning rate for everything; chooses by belief (stakes ignored).	alpha = 0.18, beta = 1.42	920
M_type_SP	8	pre-task rating	by partner type (6)	belief: $\beta(Q - 0.5)$	yes	no	alpha_TC, alpha_TI, alpha_UC, alpha_UI, alpha_TA, alpha_UA, beta, phi	Starts at the pre-task stereotype rating; a separate learning rate for each of the 6 partner types; chooses by belief (stakes ignored); tends to repeat its last choice.	alpha_TC = 0.09, alpha_TI = 0.07, alpha_UC = 0.06, alpha_UI = 0.11, alpha_TA = 0.08, alpha_UA = 0.12, beta = 3.54, phi = 0.41	931

Model	k	Starting belief	Learning rate	Choice rule	Stickiness	β by last outcome	Free parameters	What it assumes	Median fitted values	Δ BIC (summed)
M_type_S	7	pre-task rating	by partner type (6)	belief: $\beta(Q - 0.5)$	no	no	alpha_TC, alpha_TI, alpha_UC, alpha_UI, alpha_TA, alpha_UA, beta	Starts at the pre-task stereotype rating; a separate learning rate for each of the 6 partner types; chooses by belief (stakes ignored).	alpha_TC = 0.10, alpha_TI = 0.06, alpha_UC = 0.05, alpha_UI = 0.11, alpha_TA = 0.08, alpha_UA = 0.12, beta = 4.70	953
M1v	2	0.5	single	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	no	no	alpha, beta	Starts at 0.5; one learning rate for everything; chooses by expected value (stakes matter).	alpha = 0.27, beta = 0.09	1000
M_cong_Sv	4	pre-task rating	by congruence (3)	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	no	no	alpha_c, alpha_i, alpha_amb, beta	Starts at the pre-task stereotype rating; separate learning rates for congruent, incongruent, and ambiguous partners; chooses by expected value (stakes matter).	alpha_c = 0.09, alpha_i = 0.06, alpha_amb = 0.07, beta = 0.34	1059
M_cong	4	0.5	by congruence (3)	belief: $\beta(Q - 0.5)$	no	no	alpha_c, alpha_i, alpha_amb, beta	Starts at 0.5; separate learning rates for congruent, incongruent, and ambiguous partners; chooses by belief (stakes ignored).	alpha_c = 0.11, alpha_i = 0.01, alpha_amb = 0.01, beta = 4.97	1113
M_type_P	8	0.5	by partner type (6)	belief: $\beta(Q - 0.5)$	yes	no	alpha_TC, alpha_TI, alpha_UC, alpha_UI, alpha_TA, alpha_UA, beta, phi	Starts at 0.5; a separate learning rate for each of the 6 partner types; chooses by belief (stakes ignored); tends to repeat its last choice.	alpha_TC = 0.10, alpha_TI = 0.00, alpha_UC = 0.02, alpha_UI = 0.02, alpha_TA = 0.01, alpha_UA = 0.00, beta =	1149

Model	k	Starting belief	Learning rate	Choice rule	Stickiness	β by last outcome	Free parameters	What it assumes	Median fitted values	Δ BIC (summed)
									10.31, $\phi = 0.40$	
M_type_SPv	8	pre-task rating	by partner type (6)	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	yes	no	α_{TC} , α_{TI} , α_{UC} , α_{UI} , α_{TA} , α_{UA} , β , ϕ	Starts at the pre-task stereotype rating; a separate learning rate for each of the 6 partner types; chooses by expected value (stakes matter); tends to repeat its last choice.	$\alpha_{TC} = 0.10$, $\alpha_{TI} = 0.04$, $\alpha_{UC} = 0.00$, $\alpha_{UI} = 0.24$, $\alpha_{TA} = 0.03$, $\alpha_{UA} = 0.11$, $\beta = 0.38$, $\phi = 0.48$	1250
M_type	7	0.5	by partner type (6)	belief: $\beta(Q - 0.5)$	no	no	α_{TC} , α_{TI} , α_{UC} , α_{UI} , α_{TA} , α_{UA} , β	Starts at 0.5; a separate learning rate for each of the 6 partner types; chooses by belief (stakes ignored).	$\alpha_{TC} = 0.10$, $\alpha_{TI} = 0.00$, $\alpha_{UC} = 0.03$, $\alpha_{UI} = 0.02$, $\alpha_{TA} = 0.01$, $\alpha_{UA} = 0.00$, $\beta = 14.45$	1292
M_congv	4	0.5	by congruence (3)	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	no	no	α_c , α_i , α_{amb} , β	Starts at 0.5; separate learning rates for congruent, incongruent, and ambiguous partners; chooses by expected value (stakes matter).	$\alpha_c = 0.42$, $\alpha_i = 0.18$, $\alpha_{amb} = 0.30$, $\beta = 0.13$	1311
M_type_Sv	7	pre-task rating	by partner type (6)	expected value: $\beta \cdot \Delta a(1.5Q - 1)$	no	no	α_{TC} , α_{TI} , α_{UC} , α_{UI} , α_{TA} , α_{UA} , β	Starts at the pre-task stereotype rating; a separate learning rate for each of the 6 partner types; chooses by expected value (stakes matter).	$\alpha_{TC} = 0.04$, $\alpha_{TI} = 0.04$, $\alpha_{UC} = 0.00$, $\alpha_{UI} = 0.22$, $\alpha_{TA} = 0.01$, $\alpha_{UA} = 0.15$, $\beta = 0.41$	1455

Parameters

Parameter	Meaning	Range	Used in
alpha	Learning rate: how far belief moves toward each outcome	0 to 1	M1, M3, M5, M7, M1v, M3v, M5v, M7v
beta	Choice consistency: how strongly belief drives the choice	0.001 to 5 (value); 0.01 to 20 (belief)	M1, M2, M3, M3b, M5, M6, M7, M7b, M_cong, M_cong_P, M_cong_S, M_cong_SP, M1v, M2v, M3v, M3bv, M5v, M6v, M7v, M7bv, M_congv, M_cong_Sv, M_cong_SPv, M_type, M_type_P, M_type_S, M_type_SP, M_type_Sv, M_type_SPv
alpha_r	Learning rate after the partner reciprocates	0 to 1	M2, M3b, M4, M6, M7b, M8, M2v, M3bv, M6v, M7bv
alpha_d	Learning rate after the partner defects	0 to 1	M2, M3b, M4, M6, M7b, M8, M2v, M3bv, M6v, M7bv
phi	Stickiness: tendency to repeat the last choice with this partner	-5 to 5	M3, M3b, M4, M7, M7b, M8, M_cong_P, M_cong_SP, M3v, M3bv, M7v, M7bv, M_cong_SPv, M_type_P, M_type_SP, M_type_SPv
beta_r	Choice consistency when the last outcome was reciprocation (or no feedback yet)	0.01 to 20	M4, M8
beta_d	Choice consistency when the last outcome was a defection	0.01 to 20	M4, M8
alpha_c	Learning rate for congruent partners (behavior fits the stereotype)	0 to 1	M_cong, M_cong_P, M_cong_S, M_cong_SP, M_congv, M_cong_Sv, M_cong_SPv
alpha_i	Learning rate for incongruent partners (behavior violates the stereotype)	0 to 1	M_cong, M_cong_P, M_cong_S, M_cong_SP, M_congv, M_cong_Sv, M_cong_SPv
alpha_amb	Learning rate for ambiguous partners (return 50%)	0 to 1	M_cong, M_cong_P, M_cong_S, M_cong_SP, M_congv, M_cong_Sv, M_cong_SPv
alpha_TC	Learning rate for the trustworthy occupation, returns 80% (congruent) partner	0 to 1	M_type, M_type_P, M_type_S, M_type_SP, M_type_Sv, M_type_SPv
alpha_TI	Learning rate for the trustworthy occupation, returns 20% (incongruent) partner	0 to 1	M_type, M_type_P, M_type_S, M_type_SP, M_type_Sv, M_type_SPv
alpha_UC	Learning rate for the untrustworthy occupation, returns 20% (congruent) partner	0 to 1	M_type, M_type_P, M_type_S, M_type_SP, M_type_Sv, M_type_SPv

Parameter	Meaning	Range	Used in
alpha_UI	Learning rate for the untrustworthy occupation, returns 80% (incongruent) partner	0 to 1	M_type, M_type_P, M_type_S, M_type_SP, M_type_Sv, M_type_SPv
alpha_TA	Learning rate for the trustworthy occupation, returns 50% (ambiguous) partner	0 to 1	M_type, M_type_P, M_type_S, M_type_SP, M_type_Sv, M_type_SPv
alpha_UA	Learning rate for the untrustworthy occupation, returns 50% (ambiguous) partner	0 to 1	M_type, M_type_P, M_type_S, M_type_SP, M_type_Sv, M_type_SPv